

80%

→ sorting
→ DP
→

TC, SC
Algo

Lec_58

Introduction to greedy Algorithms

1-2

1-2

• greedy = optimal

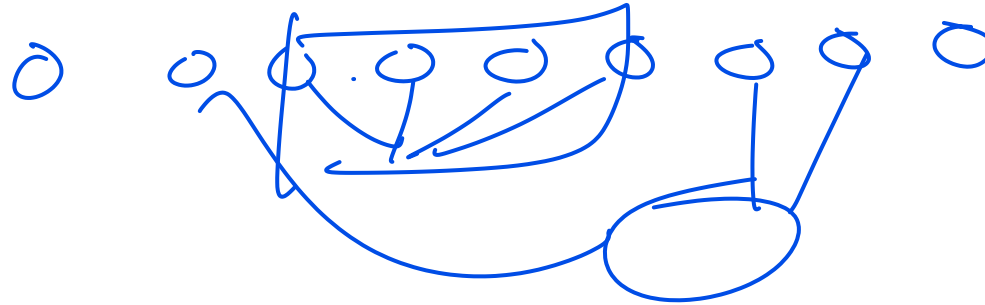


The Greedy method is the simplest and straightforward approach. It is not an algorithm, but it is a technique

- A greedy algorithm is an approach for solving a problem by selecting the best option available at the moment. It doesn't worry whether the current best result will bring the overall optimal result.

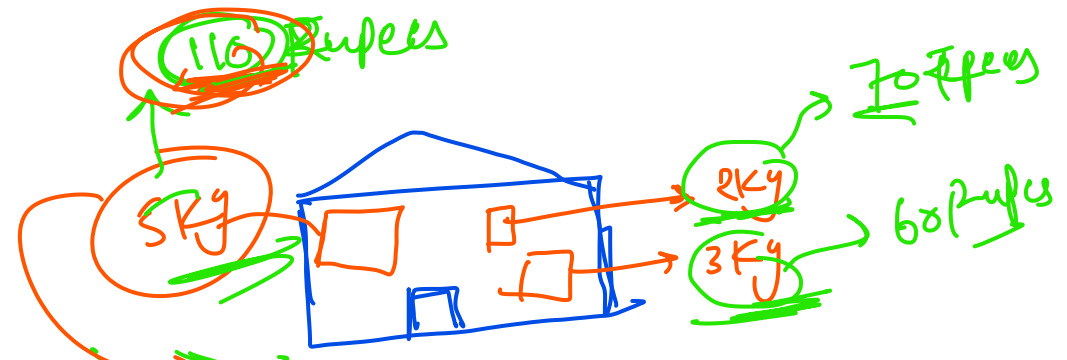
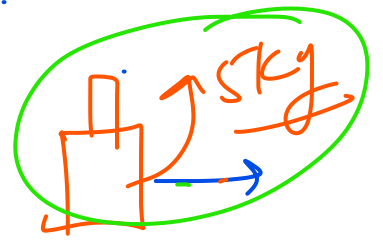
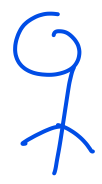
- Most of the Problems in greedy contain n-numbers of inputs and our requirement is finding a subset which will satisfy our conditions and optimizes our goal

- Ex:



Ex:

ek chor $\frac{4}{7}$

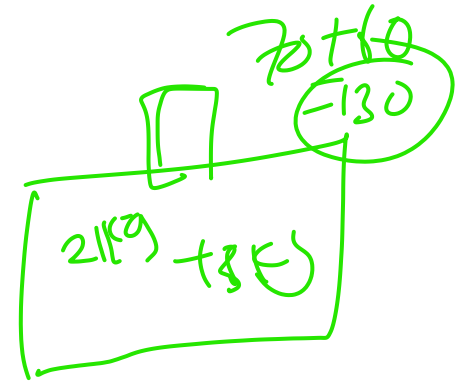
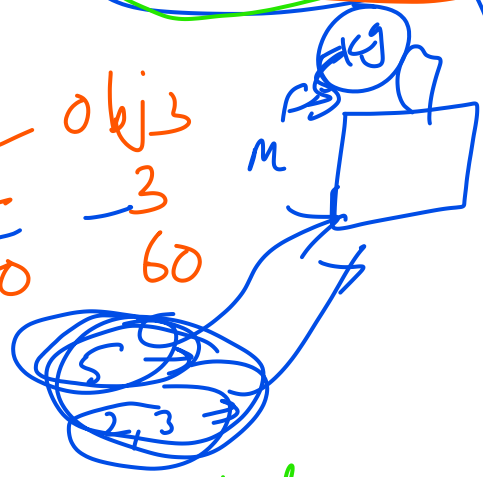


input

3 object

W
P

obj1	obj2	obj3
5	2	3
<u>100</u>	<u>70</u>	<u>60</u>



greed knapsack problem

Basic of Greedy :

• Solution Space : Set of all possible Solutions over given n-inputs

• Feasible Solution : Set of all Solutions which will satisfy our conditions

• Optimal Solution : It is one of the feasible Solution which optimized our goal, it need not be unique

~~5, 2, 2~~

5, 1, 9
5, 2, 3, 1, 9
3, 7, 2

~~5, 2~~
~~2, 5~~

5, 1, 9

2, 3, 1, 9

2, 3, 1, 9

→ main problem

Applications of Greedy (in GATE):

- Knapsack Problem ✓
- Job Sequencing With Deadlock ✓
- Huffman Coding ✓
- Optimal Merge Pattern ✓
- Minimum Spanning tree ✓

Homework

5

①

Greedy

Knapsack

Problem

